

4.4 Practice - Three Variables

Solve each of the following systems of equation.

1) $a - 2b + c = 5$
 $2a + b - c = -1$
 $3a + 3b - 2c = -4$

2) $2x + 3y = z - 1$
 $3x = 8z - 1$
 $5y + 7z = -1$

3) $3x + y - z = 11$
 $x + 3y = z + 13$
 $x + y - 3z = 11$

4) $x + y + z = 2$
 $6x - 4y + 5z = 31$
 $5x + 2y + 2z = 13$

5) $x + 6y + 3z = 4$
 $2x + y + 2z = 3$
 $3x - 2y + z = 0$

6) $x - y + 2z = -3$
 $x + 2y + 3z = 4$
 $2x + y + z = -3$

7) $x + y + z = 6$
 $2x - y - z = -3$
 $x - 2y + 3z = 6$

8) $x + y - z = 0$
 $x + 2y - 4z = 0$
 $2x + y + z = 0$

9) $x + y - z = 0$
 $x - y - z = 0$
 $x + y + 2z = 0$

10) $x + 2y - z = 4$
 $4x - 3y + z = 8$
 $5x - y = 12$

11) $-2x + y - 3z = 1$
 $x - 4y + z = 6$
 $4x + 16y + 4z = 24$

12) $4x + 12y + 16z = 4$
 $3x + 4y + 5z = 3$
 $x + 8y + 11z = 1$

13) $2x + y - 3z = 0$
 $x - 4y + z = 0$
 $4x + 16y + 4z = 0$

14) $4x + 12y + 16z = 0$
 $3x + 4y + 5z = 0$
 $x + 8y + 11z = 0$

15) $3x + 2y + 2z = 3$
 $x + 2y - z = 5$
 $2x - 4y + z = 0$

16) $p + q + r = 1$
 $p + 2q + 3r = 4$
 $4p + 5q + 6r = 7$

17) $x - 2y + 3z = 4$
 $2x - y + z = -1$
 $4x + y + z = 1$

18) $x + 2y - 3z = 9$
 $2x - y + 2z = -8$
 $3x - y - 4z = 3$

19) $x - y + 2z = 0$
 $x - 2y + 3z = -1$
 $2x - 2y + z = -3$

20) $4x - 7y + 3z = 1$
 $3x + y - 2z = 4$
 $4x - 7y + 3z = 6$

21) $4x - 3y + 2z = 40$
 $5x + 9y - 7z = 47$
 $9x + 8y - 3z = 97$

22) $3x + y - z = 10$
 $8x - y - 6z = -3$
 $5x - 2y - 5z = 1$

$$\begin{aligned} 23) \quad & 3x + 3y - 2z = 13 \\ & 6x + 2y - 5z = 13 \\ & 5x - 2y - 5z = -1 \end{aligned}$$

$$\begin{aligned} 25) \quad & 3x - 4y + 2z = 1 \\ & 2x + 3y - 3z = -1 \\ & x + 10y - 8z = 7 \end{aligned}$$

$$\begin{aligned} 27) \quad & m + 6n + 3p = 8 \\ & 3m + 4n = -3 \\ & 5m + 7n = 1 \end{aligned}$$

$$\begin{aligned} 29) \quad & -2w + 2x + 2y - 2z = -10 \\ & w + x + y + z = -5 \\ & 3w + 2x + 2y + 4z = -11 \\ & w + 3x - 2y + 2z = -6 \end{aligned}$$

$$\begin{aligned} 31) \quad & w + x + y + z = 2 \\ & w + 2x + 2y + 4z = 1 \\ & -w + x - y - z = -6 \\ & -w + 3x + y - z = -2 \end{aligned}$$

$$\begin{aligned} 24) \quad & 2x - 3y + 5z = 1 \\ & 3x + 2y - z = 4 \\ & 4x + 7y - 7z = 7 \end{aligned}$$

$$\begin{aligned} 26) \quad & 2x + y = z \\ & 4x + z = 4y \\ & y = x + 1 \end{aligned}$$

$$\begin{aligned} 28) \quad & 3x + 2y = z + 2 \\ & y = 1 - 2x \\ & 3z = -2y \end{aligned}$$

$$\begin{aligned} 30) \quad & -w + 2x - 3y + z = -8 \\ & -w + x + y - z = -4 \\ & w + x + y + z = 22 \\ & -w + x - y - z = -14 \end{aligned}$$

$$\begin{aligned} 32) \quad & w + x - y + z = 0 \\ & -w + 2x + 2y + z = 5 \\ & -w + 3x + y - z = -4 \\ & -2w + x + y - 3z = -7 \end{aligned}$$